

9. Interface 1KB memory using NAND gate, with address range 1000H to 13FFFH.

Solution,

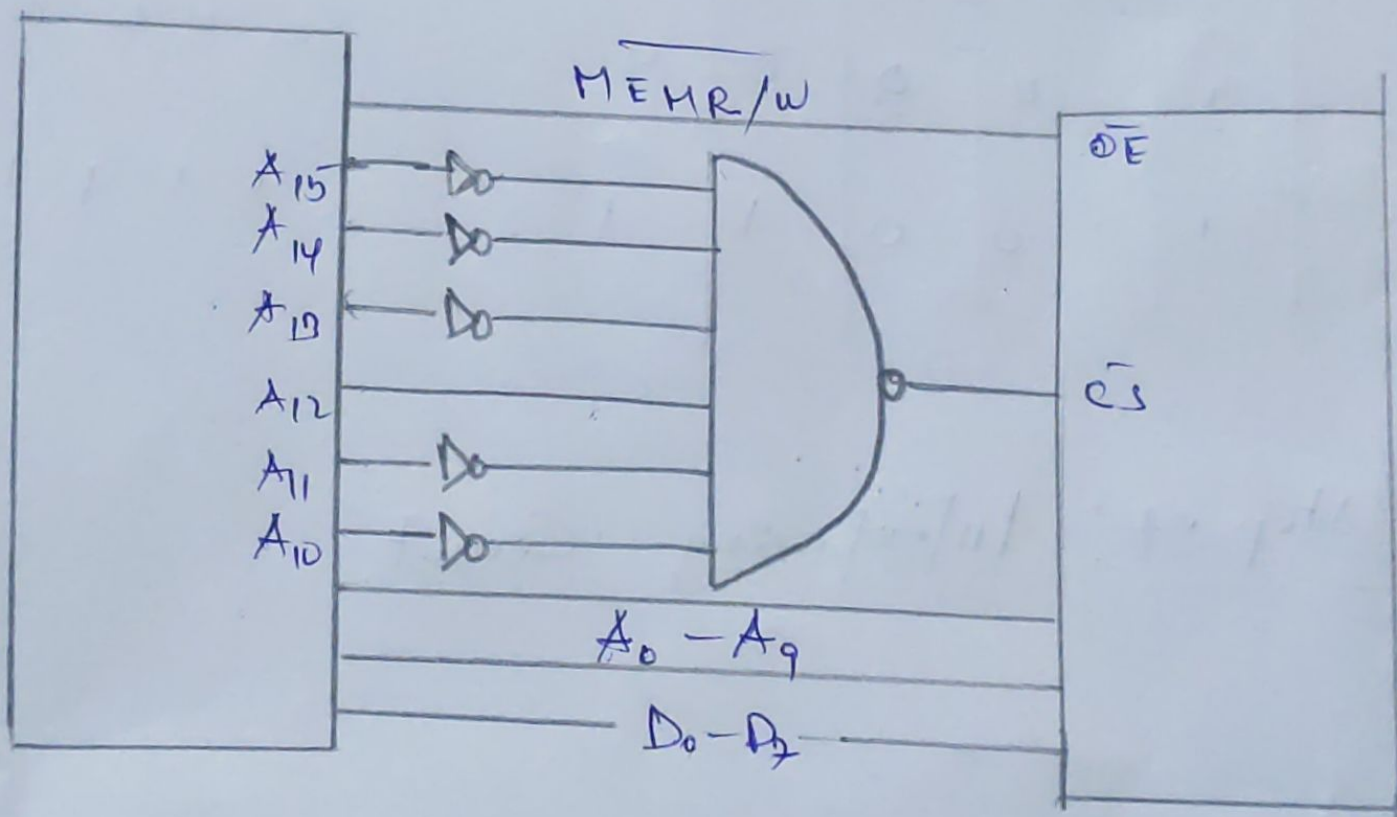
$$\begin{aligned} \text{Address line} &= 1\text{KB} \times 2^{10} \\ &= 2^0 \times 10 \times 2^{10} \end{aligned}$$

$$\text{Address line} = \underline{\underline{20}}$$

Consider

A_{15}	A_{14}	A_{13}	A_{12}	A_{11}	A_{10}	A_9	A_8	...	A_3	A_2	A_1	A_0	
0	0	0	1	0	0	0	0		0	0	0	0	1000H
0	0	0	1	0	0	1	1		1	1	1	1	13FFFH

They interface will be.



10 Interface two $8K \times 8$ RAM and a $8K \times 8$ EPROM chip with the 8085, using 74LS138 decoder, such that the starting address assigned to them are $6000H$, $8000H$ and $0000H$ respectively.

Answer:

Step 01: Determine address line for each.

$$\begin{aligned} &\underline{8K \times 8 \text{ RAM}} \\ &= 8 \times 2^{10} \\ &= 2^3 \times 2^{10} \\ &= 2^{13} \\ &= 13 \text{ address line.} \end{aligned}$$

$$\begin{aligned} &\underline{8K \times 8 \text{ EPROM}} \\ &= 8 \times 2^{10} \\ &= 2^3 \times 2^{10} \\ &= 2^{13} \\ &= 13 \text{ address line.} \end{aligned}$$

Step 02: Address table.

	A_{15}	A_{14}	A_{13}	A_{12}	A_{11}	A_{10}	A_9	A_8	A_7	A_6	A_5	A_4	A_3	A_2	A_1	A_0	Addresses.
RAM	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	$6000H$
	0	1	1	1	1	0	0	0	0	0	0	0	1	1	1	1	$7FFFH$
EPROM	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	$8000H$
	1	0	0	1	1	0	0	0	0	0	0	0	1	1	1	1	$9FFFH$

Step 04: Interfacing Circuit.

10. Interfacing circuit

